

Matteo Romilio Pasqual

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”In the middle of difficulty lies opportunity.” – Albert Einstein

🎓 Education

Politecnico di Milano (polimi.it 🔗)	<i>Sep 2024 – Jul 2026</i>
M.Sc. in Computer Science and Engineering - Artificial intelligence	Grade: 110L/110
Politecnico di Milano (polimi.it 🔗)	<i>Sep 2021 – Jul 2024</i>
B.Sc. in Computer Science and Engineering	Grade: 109/110
Liceo Scientifico ”Leonardo da Vinci” (liceogallarate.it 🔗)	<i>Sep 2016 – Jul 2021</i>
Diploma di Liceo Scientifico - Scienze Applicate	Grade: 96/100

🔬 Projects and Research

M-SHRED: Masked Shallow Recurrent Decoders for Fault-Tolerant Sparse Sensing under Missing Sensor Measurements

Master Thesis - 2026

- Designed and validated a hierarchy of Transformer-based masking mechanisms to make sparse-sensing reconstruction models explicitly aware of missing sensor data, and introduced a multi-step decoding training strategy that further improved reconstruction accuracy and latent trajectory smoothness, with no additional inference-time overhead
- Gained hands-on research experience in deep learning and reduced-order modeling, alongside strengthened skills in experimental design, scientific writing, and end-to-end reproducible ML research on both synthetic and real-world noisy datasets

Multi-agent System Leonardo Drone Contest 2025

Challenge - 2025

- Ranked 2nd nationally in a robotics competition ([LDC-2025](#) [🔗](#)) among teams from top Italian universities
- Developed planning, exploration, and localization algorithms for a collaborative rover-drone system, together with computer vision modules for ArUco/QR marker detection and unknown object detection, to dynamically execute and reschedule mission tasks in real time based on external triggers
- Achieved reliable autonomous target and object detection, marker-sequence following, and coordinated rover-drone task hand-off, validating the robustness of the proposed planning and perception pipeline under dynamic, real-time mission constraints

Course Projects

Projects - 2025

- Artificial Neural Networks and Deep Learning (ANNDL) course project: multivariate time-series classification using neural networks ([GitHub](#) [🔗](#))
- Artificial Neural Networks and Deep Learning (ANNDL) course project: biomedical image classification using deep learning methods ([GitHub](#) [🔗](#))
- Mathematical Models and Methods for Image Processing (MMMIP) course project: image processing for denoising, inpainting, and anomaly detection ([GitHub](#) [🔗](#))
- Online Learning Applications (OLA) course project: pricing under budget constraints using online learning methods ([GitHub](#) [🔗](#))

Kolmogorov-Arnold Networks (KANs) lecture

Project - 2024

- Studied and presented Kolmogorov-Arnold Networks (KANs), a spline-based alternative to MLPs inspired by the Kolmogorov-Arnold theorem, as part of the Numerical Analysis for Machine Learning course
- Analyzed KANs’ theoretical foundations and architectural differences from MLPs, covering learnable B-spline activation functions, training dynamics, and advanced techniques (sparsification, pruning, symbolic regression) for improved interpretability
- Implemented and benchmarked KANs on regression and real-world classification tasks, achieving a higher classification accuracy, and published the code and materials ([GitHub](#) [🔗](#))

Non-intrusive Monitoring of Daily Habits in Older Adults:

Research - 2024

A Chaos Game Representation-Based Approach

- Published a short paper at UCAmI 2025 – 17th International Conference on Ubiquitous Computing and Ambient Intelligence ([Paper](#) [🔗](#))
- Developed a novel framework using Chaos Game Representation (CGR) to transform time-series sensor data into fractal-like images for behavioural analysis, enabling classification of daily routines
- Demonstrated effective anomaly detection and clustering of behavioural patterns on both synthetic and real-world multi-resident datasets

📄 Publications

Non-intrusive Monitoring of Daily Habits in Older Adults:

A Chaos Game Representation-Based Approach

UCAmI 2025 – 17th International Conference on Ubiquitous Computing and Ambient Intelligence ([DOI](#) [🔗](#))

👤 Experience

University Multichance Tutor

Milano, IT

Politecnico di Milano

Sep 2025 – Jul 2026

- Provided one-to-one academic tutoring to university students through the Multichance program
- Provided personalized academic support and study assistance

University Peer-to-peer Tutor

Milano, IT

Politecnico di Milano

Sep 2022 – Jul 2024

- Provided one-to-one academic tutoring to university students through the Peer-to-Peer tutoring program
- Provided personalized academic support and study assistance

🌐 Languages

- **Italian:** Native
- **English:** Upper Intermediate (B2+)

🔧 Skills

- **AI & Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, OpenCV
- **Programming & Software Dev.:** Python, C/C++, Java
- **Databases Technologies:** SQL, MongoDB, Spark, Neo4j, Zilliz-Milvus, Pandas
- **OS & Middleware:** ROS, ROS2, Linux, Docker and VMs
- **Other Technologies:** Matlab, VHDL/Vivado, Git/GitHub, L^AT_EX

🏆 Awards

Best Freshmen Award: top 5% freshmen of PoliMi

2023

👉 Volunteering

Volunteer at Oratorio di Albizzate

Sep 2016 - Aug 2021

Albizzate

- Organized educational, recreational, and community activities for children and teenagers